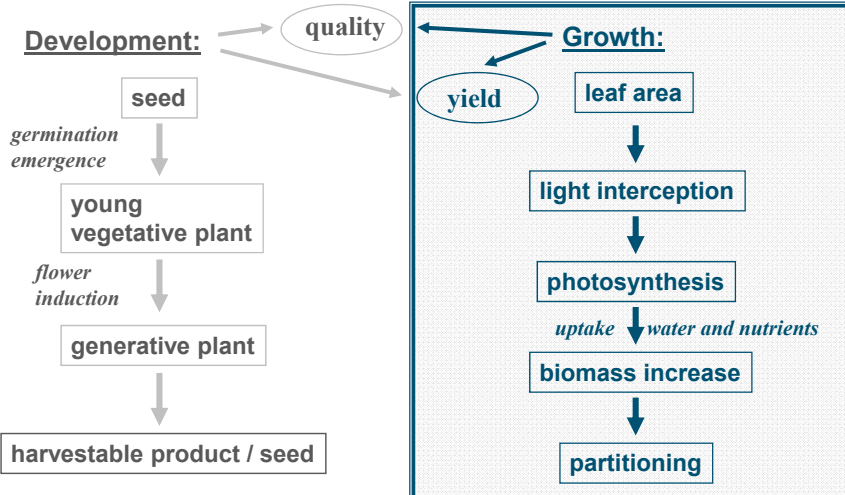


Lintul2: Water limited crop growth

Dr. ir. Peter Leffelaar - Wageningen University

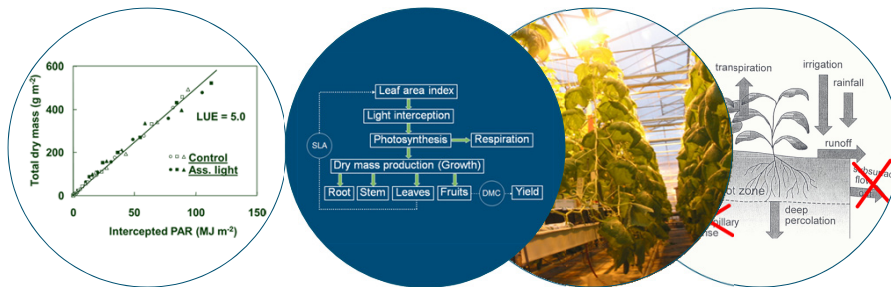


Lintul1: Water limited crop growth - scope of the course -

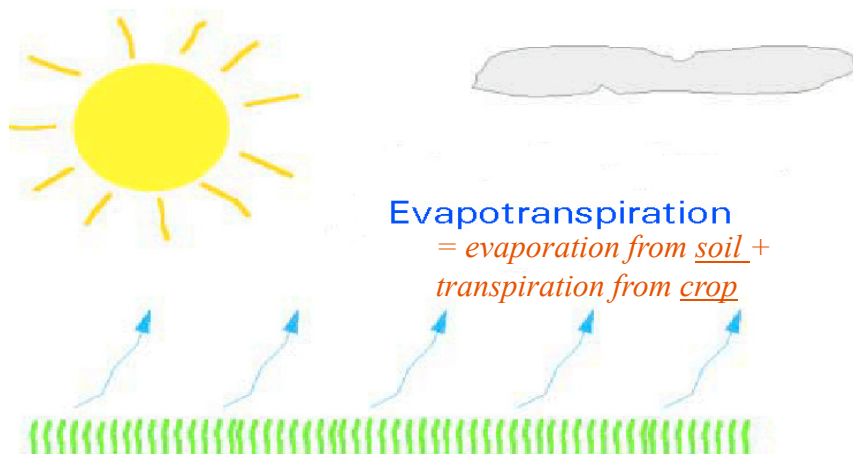


Lintul2: Water limited crop growth

What drives Evapotranspiration?

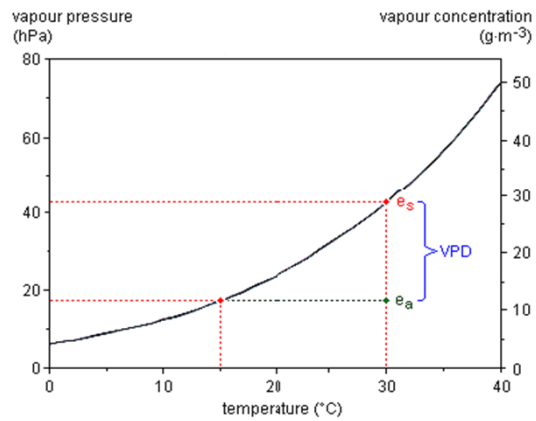


Evapotranspiration



Temperature and saturation vapour pressure

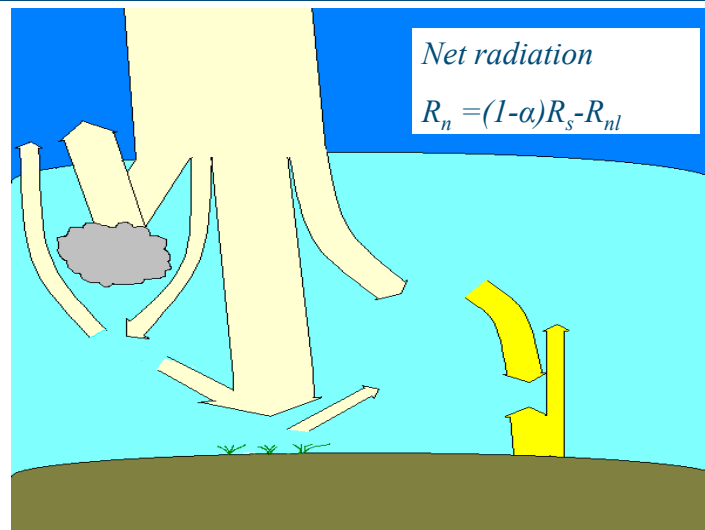
The relationship between Temperature and Saturation Vapour Pressure / Concentration



$$e_s = 6.11 e^{\{(17.47 T_s) / (239 + T_s)\}}$$

$$\Delta = \frac{4175.3 \left(6.11 e^{\{(17.47 T_s) / (239 + T_s)\}} \right)}{(239 + T_s)^2}$$

Energy balance



How much heat is needed for evaporation of water?

$$\lambda = 2.45 \text{ MJ/kg (around } 20^\circ\text{C)} = 2.45 \text{ kJ/g}$$

Drivers of evapo(transpi)ration

- Radiation
- Evaporative demand

These are combined in the Penman and Penman-Monteith equations for estimating Potential Evapotranspiration

↓
(from an amply watered, 12 cm tall grass crop with an albedo of 0.23)=
Reference evapotranspiration

Penman (1948) equation used in Lintul2:

$$ET_0 = \frac{0.408 \Delta R_n}{\Delta + \gamma} + \frac{\gamma(e_s - e_a) * 2.63(1.0 + 0.54u_2)}{\Delta + \gamma}$$

Penman modified

1956: Monteith and Rijtema independently modified the Penman equation to include surface & aerodynamic resistances of crops

The full Penman-Monteith equation:

$$ET = \frac{\Delta R_n + \rho_a c_p \frac{e_s - e_a}{r_a}}{\Delta + \gamma \left(1 + \frac{r_s}{r_a}\right)}$$

where:

r_a is aerodynamic resistance

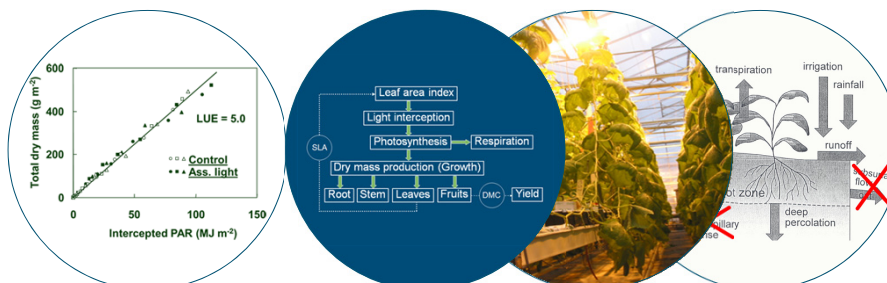
r_s is stomatal resistance

For a 12cm tall grass surface:

$$ET_0 = \frac{0.408 \Delta (R_n) + \gamma \frac{900}{T + 273} u_2 (e_s - e_a)}{\Delta + \gamma (1 + 0.34 u_2)}$$

Lintul2: Water limited crop growth

A simpler evapotranspiration method



Data needed by Penman & Penman-Monteith equations

- Solar radiation
- Temperature
- Humidity / Vapour pressure
- Wind speed

Other equations with less data needs, e.g.:

- Makkink
- Priestley-Taylor
- Blaney-Criddle
- Turc

(Even more methods exist: Hargreaves, McNaughton & Black, Linacre, Thornthwaite, Shuttleworth-Wallace, Jensen-Haise, etc.)



Turc Method (1961)

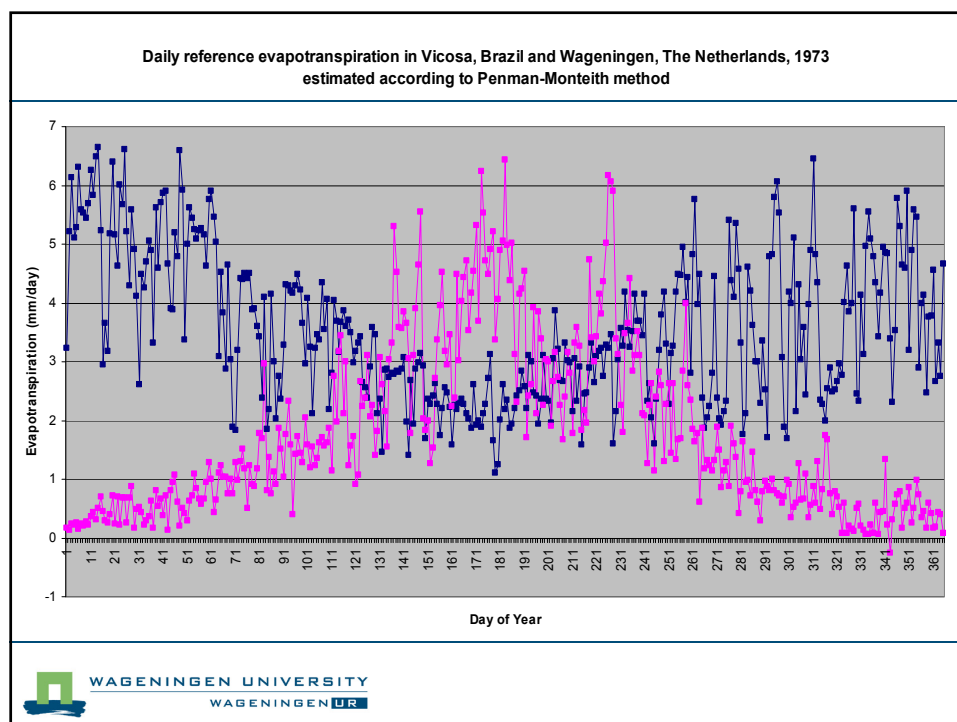
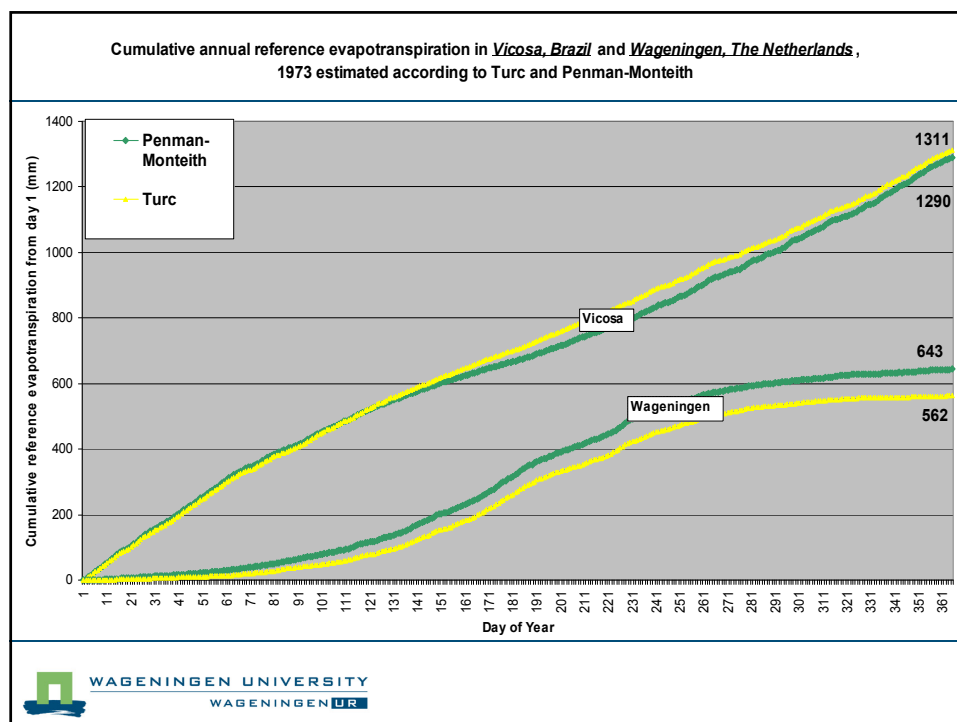
$$ET_0 = 0.013 \frac{T}{T+15} (Rs + 50)$$

(For Relative Humidity > 50%)

$$ET_0 = 0.013 \frac{T}{T+15} (Rs + 50) \left(1 + \frac{50 - RH}{70} \right)$$

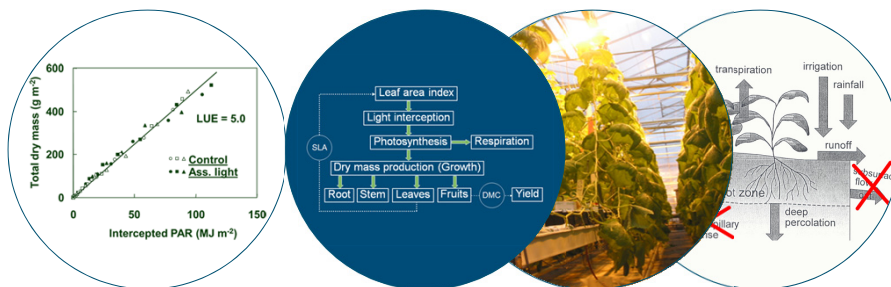
(For Relative Humidity < 50%)





Lintul2: Water limited crop growth

Coupling environment with crop growth



Coupling environmental conditions with crop growth

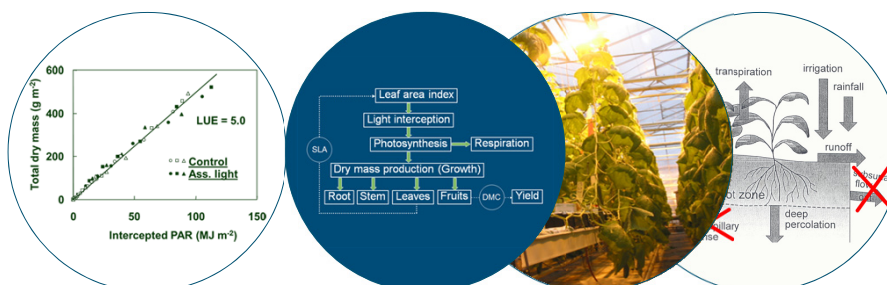
- General model ideas
- The water balance
- Effects of water on:
 - Roots
 - Emergence
 - Overall growth rate
 - Partitioning
- Transpiration reduction factor
- Relational diagram
- Practical

General modelling ideas

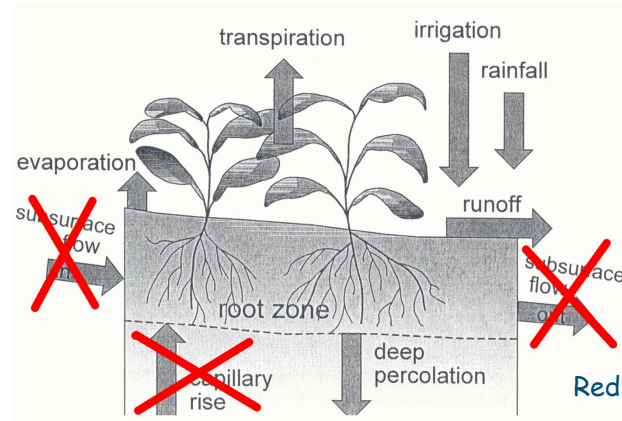
- **Crop - LINTUL1**
 - LUE based
 - Development based on TSUM
 - Exponential light extinction
 - Fixed partitioning of assimilates
- **Soil - One layer**
 - Increasing in depth with growing roots
 - Exploring new soil means new water, assuming $pF = 2$
 - Immediate homogeneous incoming water distribution
 - Water held at $pF = 2$ (field capacity)
- **Crop <-> Soil**
 - Overall crop growth affected by water shortage
 - Root - shoot partitioning affected
 - Emergence affected
 - Root growth rate affected

Lintul2: Water limited crop growth

Water or soil moisture balance



Soil water balance terms of the root zone



Red crosses:
not in LINTUL2

From: Allen, Pereira, Raes & Smith, 1998

Water dynamics in LINTUL: (1) placement of calculations

```

DEFINE_CALL GLA      (...)
DEFINE_CALL PENMAN   (...)
DEFINE_CALL EVAPTR   (...)
DEFINE_CALL DRUNIR   (...)
TITLE LINTUL2
*** 1. Initial conditions and run control
*** 2. Environmental data and temperature sum
*** 3. Leaf growth and senescence
*** 4. Light interception and total crop growth rate
*** 5. Growth rates and dry matter production of plant organs
*** 6. Soil moisture balance
*** 7. Functions and parameters for spring wheat
END
STOP
SUBROUTINE GLA      (...)
SUBROUTINE PENMAN   (...)
SUBROUTINE EVAPTR   (...)
SUBROUTINE DRUNIR   (...)
    
```

Water dynamics in LINTUL: (2) their parameters

*** 6. Soil moisture balance

```

EXPLOR = 1000. * RROOTD * WCFC
RNINTC = MIN( RAIN , 0.25*LAI )
CALL PENMAN( DAVTMP , VP , DTR , LAI , WN , RNINTC ,      PEVAP , PTRAN )
CALL EVAPTR( PEVAP , PTRAN , RROOTD , WA , WCAD , WCWP , WCFC , WCWET , WCST , ...
            TRANCO , DELT ,                                WCCR, EVAP, TRAN )
TRANRF = TRAN / NOTNUL( PTRAN )
CALL DRUNIR( RAIN , RNINTC , EVAP , TRAN , IRRIGF , DRATE , DELT , WA , RROOTD , ...
            WCFC , WCST ,                                DRAIN , RUNOFF , IRRIG )

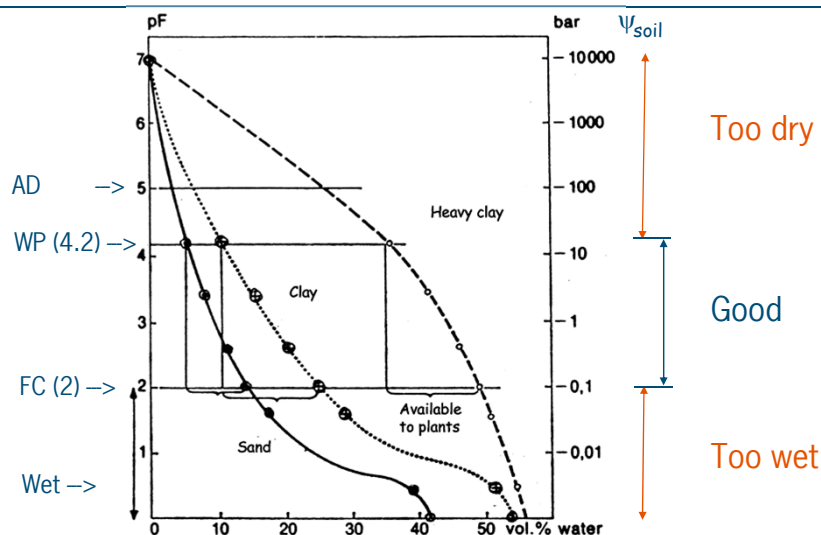
RWA = ( RAIN + EXPLOR + IRRIG ) - ( RNINTC + RUNOFF + TRAN + EVAP + DRAIN )
WA  = INTGRL( WAI , RWA )
WC  = 0.001 * WA / RROOTD
    
```

*** 7. Functions and parameters for spring wheat

```

PARAM WCI      = 0.36
PARAM RROOTDM = 1.2 ; RRDMAX = 0.012
PARAM WCAD     = 0.08 ; WCWP   = 0.23 ; WCFC = 0.36 ; WCWET = 0.48 ; WCST = 0.55
PARAM TRANCO   = 8.   ; DRATE  = 50. ; IRRIGF = 0.
    
```

Soil water retention curve



Root growth and exploration of water

TITLE LINTUL2

*** 1. Initial conditions and run control

INCON ROOTDI = 0.1

WAI = 1000. * ROOTDI * WCI

*** 2. Environmental data and temperature sum

WEATHER WTRDIR='D:\SYS\WEATHERBRAZIL\'; CNTR='BRAZIL'; ISTN=3; IYEAR=1975

* VP Vapour pressure kPa

* WN Wind speed m/s

* RAIN Precipitation mm

*** 5. Growth rates and dry matter production of plant organs

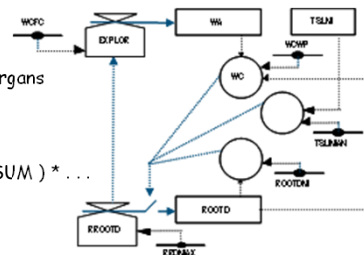
RROOTD = RRDMAX * ...

INSW(WC-WCWP, 0., 1.) * ...

REAAND(ROOTDM-ROOTD, TSUMAN-TSUM) * ...

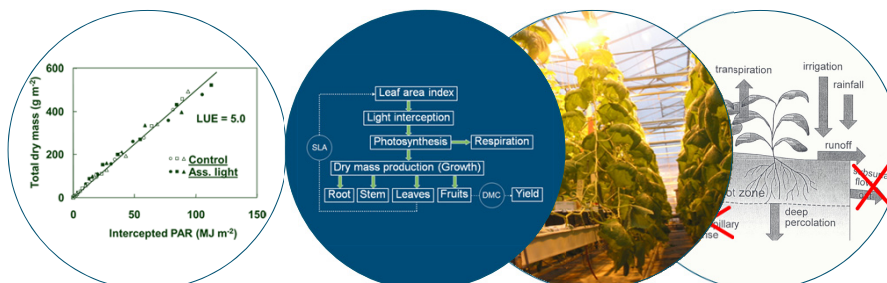
EMERG

ROOTD = INTGRL(ROOTDI, RROOTD)



Lintul2: Water limited crop growth

Emergence and crop growth are affected by water stress (TRANRF)



How water affects emergence and crop growth

TITLE LINTUL2

*** 2. Environmental data and temperature sum

```
EMERG = MAX ( REAAND(TIME-DOYEM+1.,WC-WCWP), ...
              INSW(-LAI,1,0.) )
```

*** 3. Leaf growth and senescence

```
CALL GLA( TIME , DOYEM , DTEFF , TSUM , LAII , ...
          RGRL , DELT , SLA , LAI , GLV , ...
          TRANRF , WC , WCWP ,          GLAI )
```

*** 4. Light interception and total crop growth rate

```
PARINT = 0.5 * DTR * (1. - EXP(-K*LAI))
GTOTAL = LUE * PARINT * TRANRF
```

($T_{red-tran}$, Fig.4 or FR in
Subroutine EVAPTR)



How water affects growth of leaf area

```
SUBROUTINE GLA(TIME , DOYEM , DTEFF , TSUM , LAII , RGRL , DELT , SLA , LAI , GLV ,
$              TRANRF , WC , WCWP ,          GLAI)
IMPLICIT REAL (A-Z)
```

```
GLAI = SLA * GLV
```

```
IF ( (TSUM .LT. 330.) .AND. (LAI .LT. 0.75) )
$ GLAI = LAI * (EXP(RGRL * DTEFF * DELT) - 1.) / DELT * TRANRF
```

($T_{red-tran}$, Fig.4 or FR)

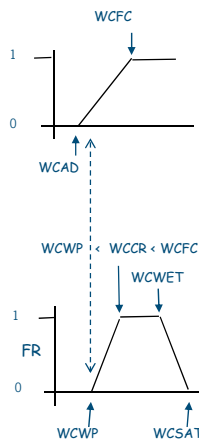
```
IF ( (TIME .GE. DOYEM) .AND. (LAI .EQ. 0.) .AND. (WC .GT. WCWP) )
$ GLAI = LAII / DELT
```

```
IF (TIME .LT. DOYEM) GLAI = 0.
```

```
RETURN
END
```



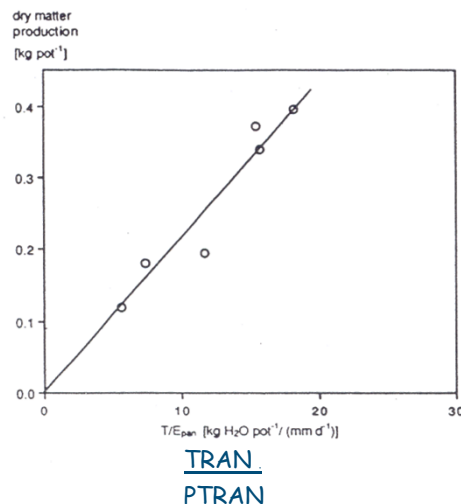
Evapo(transpi)ration reduction and physiologically determined critical point WCCR



```

SUBROUTINE EVAPTR(PEVAP ,PTRAN ,ROOTD ,WA ,WCAD ,WCWP ,WCFC ,
$                WCWET ,WCST ,TRANCO ,DELT , WCCR,EVAP,TRAN )
IMPLICIT REAL (A-Z)
WC = 0.001 * WA / ROOTD
WAAD = 1000. * WCAD * ROOTD
WAFC = 1000. * WCFC * ROOTD
{ EVAP = PEVAP * LIMIT( 0., 1., (WC-WCAD)/(WCFC-WCAD) )
  WCCR = WCWP + MAX( 0.01, PTRAN/(PTRAN+TRANCO) * (WCFC-WCWP) )
  IF (WC.GT.WCCR) THEN
    FR = LIMIT( 0., 1., (WCST-WC)/(WCST-WCWET) )
  ELSE
    FR = LIMIT( 0., 1., (WC-WCWP)/(WCCR-WCWP) )
  ENDIF
  TRAN = PTRAN * FR
  AVAILF = MIN( 1., ((WA-WAAD)/DELT)/NOTNUL(EVAP+TRAN) )
  EVAP = EVAP * AVAILF
  TRAN = TRAN * AVAILF
  RETURN
END
    
```

Experimental evidence of proportional reduction of growth with transpiration

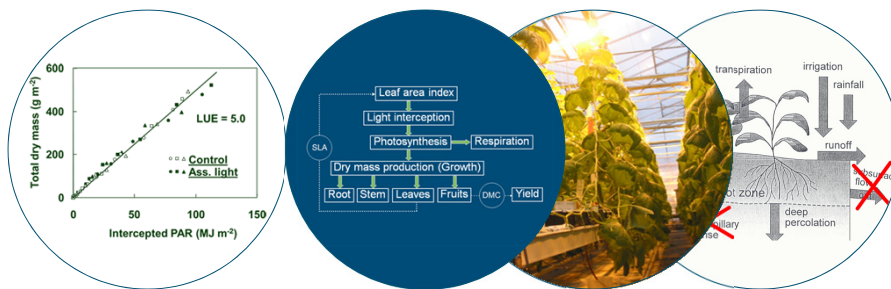


Maize in pots

From: Briggs & Shantz, 1914

Lintul2: water limited crop growth

Allocation of dry matter changes due to drought



Effect of drought on dry matter partitioning

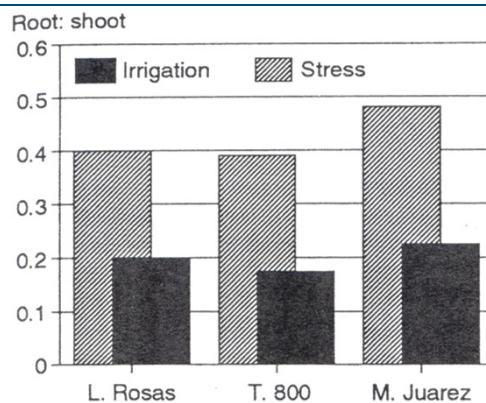
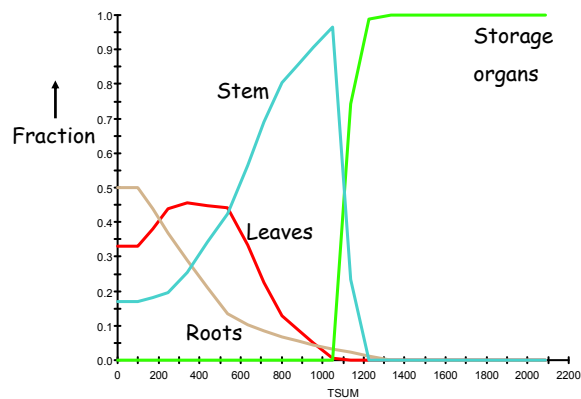


Figure 4. Root:shoot ratios in irrigated and droughted treatments. Values derive from one lysimeter per treatment.

Substrate allocation patterns in Spring Wheat



Allocation pattern of assimilates for spring wheat as used in Lintul (Light interception and utilization)

How to take account of the effect of drought on dry matter partitioning?

- The idea is:

if it is too dry,
more roots will be formed to compensate the water shortage,
and this will cost extra assimilates that go to the roots.

- Thus, less assimilates are left for the shoot, but the distribution of assimilates among the stem, the leaves and the storage organs remains the same.

Derivation of the modification factor: shmod

Idea: if too dry → more roots formed to compensate → extra assimilates to the roots (r)
Thus, less assimilates then go to the shoot, but the distribution among stem (s), leaves (l) and storage organs (so) remains the same:

$$r_{new} = r_{old} \cdot rMOD, \quad \text{with } 1 \leq rMOD \leq f_{max}$$

s + l + so then get (r_{new} - r_{old}) less assimilates in the distribution s_{old} : l_{old} : so_{old}
For instance, s will get less assimilates of quantity

$$\frac{s_{old}}{s_{old} + l_{old} + so_{old}} (r_{new} - r_{old})$$

of course s_{old} + l_{old} + so_{old} = 1 - r_{old}, so s_{new} will be: $s_{new} = s_{old} - \left\{ \frac{s_{old}}{1 - r_{old}} (r_{new} - r_{old}) \right\}$

Taking s_{old} outside brackets, the factor $\left(1 - \frac{r_{new} - r_{old}}{1 - r_{old}} \right)$ will be the sMOD or, generally, the shoot modification factor shMOD, so:

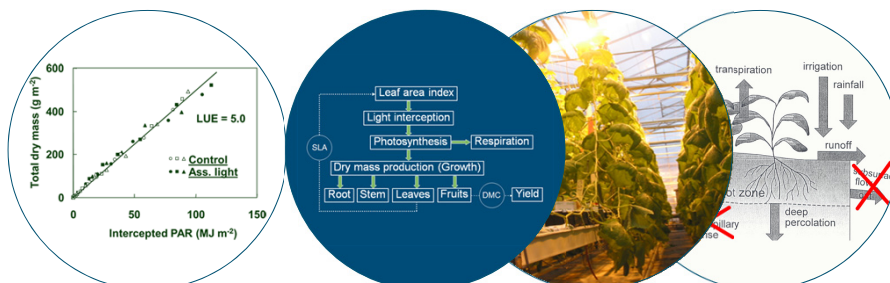
$$shMOD = 1 - \frac{r_{new} - r_{old}}{1 - r_{old}} = \frac{1 - r_{new}}{1 - r_{old}} = \frac{1 - r_{new}}{1 - \frac{r_{new}}{rMOD}}$$

where r_{old} is taken as r_{new} / rMOD according to the above 1st equation



Lintul2: Water limited crop growth

Remaining water balance terms



Drainage, runoff and irrigation

```

SUBROUTINE DRUNIR(RAIN , RNINTC , EVAP , TRAN , IRRIGF , DRATE , DELT ,
$                WA , ROOTD , WCFC , WCST ,
$                DRAIN , RUNOFF , IRRIG)
IMPLICIT REAL (A-Z)
WC  = 0.001 * WA  / ROOTD
WAFC = 1000. * WCFC * ROOTD
WAST = 1000. * WCST * ROOTD

DRAIN = LIMIT(0., DRATE, (WA-WAFC)/DELT + (RAIN - RNINTC - EVAP - TRAN) )

RUNOFF = MAX(0., (WA-WAST)/DELT + (RAIN - RNINTC - EVAP - TRAN - DRAIN) )

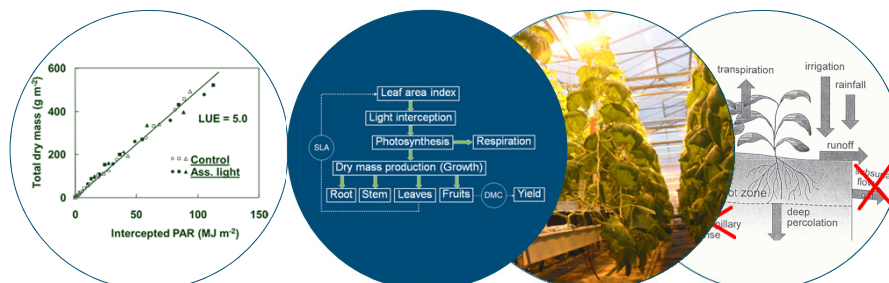
IRRIG  = IRRIGF *
        MAX(0., (WAFC-WA)/DELT - (RAIN - RNINTC - EVAP - TRAN - DRAIN - RUNOFF) )

RETURN
END

```

Lintul2: Water limited crop growth

Summarizing relational diagram



Practical exercises Lintul2

Exercises concern the following subjects:

- Differences between potential and water limited growth
- Variation of yields between different years
- Sensitivity of results for different soil water-holding capacities
- Effects of different emergence days in rain fed agriculture on yields

There are data provided for a number of years, including a dry, an average and a wet year for both Brazil and the Netherlands

At first you will do some "rule of thumb" calculations. These are then compared to the Lintul model results

Prepare a report for yourself on the PC to prepare for the exam



To be able to do the exercises, study:

- Theory: "Lintul-2: Water limited crop growth"
"Lintul-1 : Potential crop growth"
- This power point presentation



Lintul2: Water limited crop growth

References



References used in the slides

- Allen, R.G., L.S. Pereira, D. Raes & M. Smith, 1998. Crop evapotranspiration (guidelines for computing crop water requirements). FAO irrigation and drainage paper No. 56.
- Briggs L. J. & H. L. Shantz, 1914. Relative water requirements of plants. Journ. of Agr. Res. 3, 1-65.
- Koorevaar, P., G. Menelik & C. Dirksen, 1983. Elements of soil physics. Developments in soil science 13. Elsevier, Amsterdam-Oxford. pp.228.
- Magrin, G.O., A.J. Hall, S. Castellano & S.G. Meira, 1991. Rooting depth, growth cycle duration, and timing of the jointing stage in wheat: Traits that can contribute to early drought tolerance. In: 'Wheat for the Nontraditional, Warm Areas', by D.A. Saunders (Ed.), Mexico, D.F.: CIMMYT, p.509-515.
- Turc, L. (1961). Evaluation des besoins en eau d'irrigation, évapotranspiration potentielle. Ann. Agron. 12 (1): 13-49.
(Estimation of irrigation water requirements, potential evapotranspiration: A simple climatic formula evolved up to date)